

WEEK 5 PROJECT SUMMARY REPORT

1. Week 5 objective

The objective for Week 5 was to move the HealthConnect Data Science work from the problem definition established in Week 4 into the initial implementation and testing stage. The focus was on preparing the appointment dataset, investigating data quality and predictive patterns, engineering relevant features, developing a baseline classification model, evaluating the initial model outputs, and documenting limitations and next steps for Week 6.

2. Work planned

- Review Week 4 business and machine learning problem definition,
- Assess the appointment dataset for missing values, duplicates, data types, categorical consistency, and potential data quality issues,
- Define the target variable and assess variables for possible data leakage,
- Prepare the dataset for modelling while keeping the original resource separate from derived data,
- Perform exploratory data analysis to identify meaningful patterns related to appointment no-shows,
- Create engineered features where they could improve representation of potentially useful patterns,
- Define a stratified train/test strategy and develop an interpretable baseline classification model.
- Evaluate the baseline model using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC,
- Document findings, limitations, collaboration dependencies, and recommendations for Week 6.

3. Work completed

Week 5 Data Science workflow was completed through data preparation, exploratory analysis, feature engineering, feature selection, baseline modelling, and initial evaluation.

Area	Completed work
Data preparation	The prepared dataset contained 4,737 usable appointments after excluding 263 cancelled appointments. Missing value checks were

	completed, and the prepared dataset contained no missing values.
Data quality	Duplicate and categorical consistency checks were performed. A duplicate target representation was identified, and the redundant no-show column was removed.
Data leakage	Appointment outcome, waiting time, reminder information, and identifiers were excluded from the modelling predictors because they were inappropriate or potentially unavailable at the prediction point.
Exploratory data analysis (EDA)	Target distribution, numerical features, categorical features, feature target relationships, and correlations were investigated.
Feature engineering	Booking lead time was grouped into lead-time ranges and distance to the clinic was grouped into distance ranges to represent potentially non-linear patterns.
Feature selection	Twelve candidate predictors were selected based on prediction time availability, EDA findings, and potential relevance to the no-show prediction task.
Model development	A Logistic Regression pipeline was developed with standardization of numerical variables and one-hot encoding of categorical variables.
Model evaluation	The baseline model was evaluated using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC.

4. Key findings

The prepared dataset contained 4,737 appointments after cancelled appointments were excluded from the binary prediction task. The target distribution was close to being balanced: 2,423 appointments (51.15%) were no-shows, and 2,314 appointments (48.85%) were attended. The exploratory analysis identified several observable patterns relevant to the prediction task. Longer booking lead times were associated with higher numbers of no-shows. Previous no-show history also showed a clear association with the current no-show outcome. Distance to the clinic showed a tendency for the relative number of no-shows to increase as distance increased, although the most distant groups contained fewer observations. Previous appointments and previous no-shows showed the strongest relationship among the numerical predictors, with a correlation of approximately 0.461. The remaining numerical relationships were generally weak, suggesting no strong multicollinearity problem among most numerical variables examined. Potential numerical outliers were investigated using the interquartile range approach. The identified values were retained because the observed ranges were considered plausible for the appointment context and automatic removal could have discarded valid observations.

5. Baseline model results

Metric	Baseline Logistic Regression
Accuracy	61.39%
Precision	60.84%
Recall	68.87%
F1-score	64.60%
ROC-AUC	66.42%

The confusion matrix was $\begin{bmatrix} 248 & 215 \\ 151 & 334 \end{bmatrix}$. This means that 334 no-show appointments were correctly identified, while 151 actual no-shows were missed. The model also incorrectly classified 215 attended appointments as no-shows. The recall of 68.87% indicates that the baseline model identified a larger proportion of actual no-shows than it missed, but the overall performance remains moderate. The ROC-AUC of 0.6642 indicates that the model has some ability to distinguish between no-show and attended appointments, but the separation is not strong enough for the model to be considered a final or deployment-ready solution.

6. Important decisions

- Cancelled appointments were excluded before creating the binary no-show target because they do not represent the intended attended versus no-show prediction outcome,

- Identifiers such as `appointment_id` and `patient_id` were excluded because they do not provide meaningful generalizable predictive information,
- Variables that could contain information unavailable at the prediction point, including appointment outcome and waiting time, were excluded to reduce the risk of data leakage,
- One-hot encoding and numerical scaling were performed within the modelling pipeline so transformations could be fitted using the training data,
- A stratified 80/20 train/test split with `random_state=42` was used to preserve the target class distribution in both subsets,
- Logistic Regression was selected as the baseline because it is appropriate for binary classification, efficient to train, and relatively interpretable,
- Recall was considered alongside accuracy, precision, F1-score, and ROC-AUC because identifying actual no-show appointments is an important part of the prediction objective.

7. **Challenges encountered**

- Determining which variables were appropriate predictors required careful consideration of the point at which a prediction would be made,
- Potential data leakage had to be considered for variables describing outcomes or processes occurring after or around the appointment,
- Feature engineering required balancing potentially useful patterns with the need to keep the baseline model understandable,
- Initial model performance was moderate, indicating that further modelling work is required rather than treating the baseline as a final solution.

8. **Cross-Track collaboration / dependency**

A meaningful dependency was identified with the Data Analytics track. Analytical findings from the appointment dataset can inform the Data Science workflow by helping identify relevant patterns, variables, and KPIs for the prediction task. In this Week 5 workflow, findings related to booking lead time, previous no-show history, distance to the clinic, appointment type, and appointment timing were considered when selecting variables and developing engineered features. This dependency helped connect data exploration with machine learning development and ensured that modelling decisions remained linked to the wider HealthConnect problem. The collaboration or dependency also reinforced the need to distinguish information available before an appointment from information that becomes available afterwards. This supported the exclusion of potentially leaky variables and contributed to a more realistic prediction setup.

9. **Limitations, assumptions, risks and dependencies**

- The baseline Logistic Regression model provides an initial benchmark and should not be interpreted as a final production model,

- Observed associations in the exploratory analysis do not establish causation,
- Some feature groups, particularly the most distant distance group and higher previous-no-show counts, contain relatively few observations and should therefore be interpreted cautiously,
- The dataset may not fully represent all healthcare settings, patient populations, or appointment systems, so generalization to other settings would require validation,
- The model's moderate performance indicates that additional feature development, model comparison, and validation are required,
- Future work depends on maintaining clear prediction time definitions, suitable data quality, and alignment with the broader HealthConnect solution.

10. Remaining work

- Compare the Logistic Regression baseline with additional suitable classification algorithms,
- Investigate model coefficients or other feature-importance approaches to better understand predictive contributions,
- Perform additional validation and model tuning after establishing the baseline,
- Further investigate threshold selection, particularly if the practical objective prioritizes identifying more actual no-show appointments,
- Review whether additional meaningful pre-appointment features can be engineered without introducing leakage,
- Continue cross-track alignment as the HealthConnect solution develops.

11. Proposed focus for Week 6

Week 6 should focus on improving and validating the baseline model rather than immediately moving to deployment. The next stage should include comparison of alternative classification models, appropriate model tuning and validation, deeper interpretation of predictive patterns, and consideration of decision thresholds. The Data Science work should continue to remain aligned with the practical HealthConnect objective and requirements from the other project tracks.

12. Conclusion

Week 5 successfully moved the HealthConnect Data Science work from Week 4 problem definition into initial model development and testing. The appointment dataset was prepared, potential leakage was assessed, relevant features were investigated and engineered, and a Logistic Regression baseline was developed and evaluated. The baseline results show that the model has a useful but limited ability to identify potential no-show appointments, with 61.39% accuracy, 68.87% recall, and a ROC-AUC of 66.42%. The findings provide a clear benchmark for further development and highlight the need for model comparison, validation, and refinement during Week 6. Overall, the Week 5 work establishes a

structured foundation for the next phase of the HealthConnect Data Science solution while recognizing that the current model is an initial baseline rather than a final deployment ready system.